

Refine — AWS Marketplace Deployment Guide

Audience: customers who have subscribed to Blacktip Refine through AWS Marketplace and are about to deploy it in their own AWS account.

This is the **public AWS Marketplace** deployment path — a **contract-priced** subscription. You pick a tier, launch the CloudFormation stack, and Refine automatically verifies your Marketplace contract entitlement through AWS License Manager on boot. There is no license file to request, email, or upload. (Direct-sales, GovCloud / air-gapped, and Private Offer customers instead receive a signed `refine.license` ; that BYOL flow is covered in the [customer deployment guide](#).)

At a glance — the 4-step path

1. **Subscribe** to *Blacktip Refine — Multi-Cloud Cost Intelligence* in AWS Marketplace by choosing a **contract tier** (Starter / Core / Growth / Scale) for a 1-year term — or start on the **Free edition** (\$0, self-hosted, up to \$1,000/mo of analyzed cloud spend; a template-asserted Free entitlement, not a License Manager contract).
2. **Launch the CloudFormation stack** in your own AWS account (one-click from the listing). The EC2 pulls the Refine container from AWS Public ECR.
3. **Refine auto-activates**. On boot it calls **AWS License Manager**, verifies the Marketplace contract entitlement tied to your subscription, records your tier, and comes up **healthy** on its own — no license request, no emailed file, no `aws s3 cp`.
4. **Open the `RefineUrl`** from your stack Outputs, log in with the `AdminEmail` / `AdminPassword` you set as stack parameters, and add your cloud accounts.

Total elapsed: typically **10–20 minutes** from subscribe to first dashboard view — almost entirely the ~8-minute CloudFormation create.

GovCloud / air-gapped and Private Offers are different. Those customers are served through the **direct-sales BYOL** flow, not this public listing: they receive a signed `refine.license` (and, for air-gapped GovCloud, a delivery-ZIP with the image tarball, because those environments can't pull from public ECR) and drop it into a delivery bucket where an on-host watcher installs it. See the [customer deployment guide](#) for that path — none of it applies to the public Marketplace contract path on this page.

What you need before you start

Item	Why
AWS account with permission to launch CloudFormation in your target region	The stack creates EC2 and IAM resources (and, if you enable HTTPS, an ALB).
Choice of region	Where Refine will run. Closer to your cloud-account data exports = better. US East / US West are typical.
A VPC + public subnet in that region	The CFN deploys into an existing VPC. The default VPC works.
Decision: HTTPS or HTTP?	The stack launches on HTTP:8000 by default (<code>EnableHttps=false</code>) — no ACM certificate needed. Keep <code>AllowedAppCidr</code> restricted to trusted networks until you enable HTTPS. HTTPS via an ALB is an optional Day-2 upgrade (needs an ACM cert + 2 subnets in different AZs).
<code>AdminEmail</code> / <code>AdminPassword</code>	The administrator login you'll set as stack parameters and use at first sign-in.

Step 1 — Subscribe in AWS Marketplace

1. Visit the listing: <https://aws.amazon.com/marketplace/pp?sku=3o0r05v1kvpwtpr1wbnuaos7h>
2. Click **View purchase options** → choose the **contract tier** that covers your annual Spend Analyzed — Starter \$7,500 ($\leq \$100K$), Core \$20,000 ($\leq \$500K$), Growth \$50,000 ($\leq \$1M$), or Scale \$125,000 ($\leq \$5M$); larger estates use an Enterprise Private Offer. Review terms → **Accept Terms**.
3. Wait for the subscription to activate (~1 minute).
4. Click **Continue to Configuration**.

Free edition. Blacktip Refine Free is **\$0** for up to **\$1,000/month** of analyzed cloud spend — self-hosted, with a **template-asserted Free entitlement** (no AWS License Manager contract). It pauses with an upgrade prompt when analyzed spend exceeds the cap. Deploy it with the same CloudFormation stack; the paid entitlement-verification steps below don't apply on the Free path.

Refine is billed as an **annual contract through AWS Marketplace** at your chosen tier; on top of that your AWS bill reflects only the underlying EC2/EBS (and the ALB, if you enable HTTPS) the stack provisions in your own account.

Step 2 — Launch the CloudFormation stack

1. From the Configuration page, click **Launch CloudFormation**.
2. AWS opens the CloudFormation console with the Marketplace template pre-loaded.
3. Fill in the parameters:

Parameter	Recommended	Notes
Instance type	<code>t3.medium</code>	Sized for 1–10 cloud accounts. Bump to <code>m6i.xlarge</code> for 50+ accounts.
SSH key pair	(your existing key)	Required if you want SSH; optional if you'll use SSM Session Manager.
VPC + Subnet	(your existing public subnet)	Default VPC works for evaluations.
AdminEmail / AdminPassword	your chosen admin login	You'll sign in to the Refine UI with these.
Allowed CIDR for Refine UI	your office / VPN IP	In the default HTTP mode keep this restricted (not <code>0.0.0.0/0</code>). <code>0.0.0.0/0</code> is only reasonable once you're behind an ALB with HTTPS.
Enable HTTPS via ALB	<code>false</code> (default)	Launches on HTTP:8000, no ACM cert needed. Set <code>true</code> as an optional Day-2 upgrade — needs an ACM cert in the same region + 2 subnets in different AZs.
ACM certificate ARN	(only if HTTPS=true)	Only needed if HTTPS=true.
Customer name (optional)	your team / cost-center name	Applied as a tag for cost allocation.
Marketplace fulfillment mode	<code>Contract</code> (baked in)	The public-listing template ships in Contract mode — Refine verifies your entitlement via AWS License Manager. Leave as-is.
Refine image URI	(preset to <code>:latest</code>)	Default tracks the latest Refine release. To pin to a specific version, set to <code>public.ecr.aws/a4k0g9y9/refine:vx.Y.Z</code> — Watchtower won't update a pinned tag.
Enable automatic updates	<code>true</code> (default)	Deploys a Watchtower sidecar that polls AWS Public ECR for new Refine images on a schedule and rolls the container when one appears. Set to <code>false</code> to manage updates manually.
Auto-update schedule (cron, UTC)	<code>0 0 3 * * *</code> (daily 03:00 UTC)	Customer-tunable cron. Pick a window that overlaps your lowest-traffic time — updates roll the container with ~30 sec of unavailability.
Notification email for updates	(blank)	Optional. Receives a Watchtower-generated summary email when an update completes. Requires the EC2 to be able to send mail via Postfix/SMTP.
Roll back failed updates automatically	<code>true</code> (default)	Deploys a host-side monitor that tracks the digest of your last-known-healthy image. If a new image stays unhealthy (non-200 from <code>/health</code>) for >5 minutes after an auto-update, the monitor rolls the container back to the previous image and enters a 1-hour cooldown to prevent loops. Logs in <code>journalctl -fu refine-rollback-monitor.service</code> . Set to <code>false</code> to opt out.

4. Click **Next** → review → check the IAM-acknowledgement box → **Create stack**.

5. Wait for `CREATE_COMPLETE` (~5–8 minutes).

Stack Outputs — what to capture

After `CREATE_COMPLETE`, the **Outputs** tab shows everything you need. The two values that matter for the next step:

Output	What to do with it
NextStep	Read this top-line; it summarizes what to do after <code>CREATE_COMPLETE</code> .
RefineUrl	Your login URL — open it once Refine boots (Step 4).
AdminConnectMethod	How to SSH or open an SSM session if you need to log into the EC2 directly.

Step 3 — Automatic license activation (nothing to do)

There is no license request, form, or upload on the public contract path. While the stack finishes, your EC2 pulls the Refine container from AWS Public ECR, and on boot Refine calls **AWS License Manager** to verify the Marketplace contract entitlement tied to your subscription. It records your tier and the `refine` container comes up **healthy** on its own — usually within a minute of `CREATE_COMPLETE`.

You can watch it from the AWS console: **Systems Manager** → **Session Manager** → **Connect**, then `docker ps` (the `refine` container should reach `healthy`).

Direct-sales / GovCloud / Private Offer (BYOL) only. If you are *not* on this public listing, you instead receive a signed `refine.License` file (~500 bytes, Ed25519-signed) plus an activation code by email, and drop it into your delivery bucket with `aws s3 cp refine.License s3://<your-DeliveryBucketName>/refine.License`. An on-host license-watcher daemon polls the bucket every 60 seconds, atomic-swaps the file into `/opt/refine/refine.License`, restarts the container, and Refine validates the signature offline against an embedded Ed25519 public key. That flow does **not** apply to the public Marketplace contract path above — see the [customer deployment guide](#).

Step 4 — First login & cloud-account onboarding

1. Open the `RefineUrl` from Outputs (HTTP by default, or HTTPS if you enabled the ALB).
2. Log in with the **AdminEmail** / **AdminPassword** you set as stack parameters. Change the password after first login (Settings → Account).
3. Add your first cloud account from **Cloud Accounts** in the sidebar:
 - **AWS:** one-click CloudFormation per account (creates the per-account data-collection Lambda + S3 bucket).
 - **Azure:** one-line `curl` | `bash` in Azure Cloud Shell (creates Entra ID app + Reader role + Storage + Cost Management Export).
 - **GCP:** Terraform module (creates Service Account + Viewer role + BigQuery billing export).
4. Wait for the first nightly sync (typically next morning) — or click **Sync now** to backfill immediately.
5. First recommendations appear within minutes of the first sync.

Cancellation & cleanup

To stop using Refine and remove all AWS resources from your account:

1. **Unsubscribe** in AWS Marketplace (or let the subscription expire). Subscription cancellation does not stop the CFN stack.
2. **Delete the CloudFormation stack** in the AWS console (the same stack you launched in Step 2). This removes the EC2, IAM role, security groups, and the ALB (if you created one).
3. **Remove the per-account onboarding stacks** in each AWS account you added to Refine (created in Step 4) — same flow, delete from CloudFormation.
4. **Azure / GCP cleanup** (if you connected those): delete the Entra ID app / Service Account; remove the Cost Management Export / BigQuery export.

All inventory data lives in your own cloud accounts the whole time — there is no Blacktip-side data to delete because there was none to begin with.

Troubleshooting

Symptom	Likely cause	Fix
Stack stuck in <code>CREATE_IN_PROGRESS</code> >15 min	UserData step is installing Docker or pulling the Refine image from public ECR	Open SSM session, <code>tail -f /var/log/cloud-init-output.log</code>
<code>RefineUrl</code> returns 502 / connection-refused	Refine container still booting or verifying the entitlement	Wait for <code>CREATE_COMPLETE</code> plus ~1 minute; open an SSM session and run <code>docker ps</code> — the UI returns 200 once <code>refine</code> is healthy
Refine shows a "License" / entitlement error	Marketplace subscription not Active, or the EC2 role can't reach AWS License Manager	Confirm the subscription is Active in AWS Marketplace; verify the instance role allows <code>license-manager:CheckoutLicense + GetEntitlements</code> . Refine re-checks automatically
Cannot reach Refine on the configured CIDR	Security group / NACL too restrictive	Verify your office IP is in <code>AllowedAppCidr</code> ; check the SG inbound rules
Login screen says "License expired"	Your annual Marketplace contract term has ended	Renew the contract in AWS Marketplace (or contact support@blacktip-ops.com) — Refine re-checks the entitlement on renewal

For anything else, email support@blacktip-ops.com with your AWS Marketplace customer ID + stack name. Typical response within 1 business day.